



NPTEL Online Certification Course

Indian Institute of Technology Roorkee

Course Name: Power System Protection and Switchgear

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WEEK 5: ASSIGNMENT 5

Question 1:

Auto-reclosing on overhead transmission lines is justified by the statistical observation that approximately

- a) 20% of line faults are transient and 80% are permanent.
- b) 80% of line faults are transient (self-clearing) and only 20% are permanent.
- c) all line faults are permanent.
- d) no line faults are transient.

Answer: (b)

Question 2:

For high-speed auto-reclosing of EHV circuit breakers, the minimum dead time (in cycles) required for de-ionisation of the arc path is given by $t = 10.5 + (kV/34.5)$, where kV is the system line voltage. Determine the minimum dead time, in seconds, for a 220 kV system at 50 Hz.

- a) 0.21 s
- b) 0.27 s
- c) 0.34 s
- d) 0.42 s

Answer: (c)

Solution:

Using $t = 10.5 + (kV / 34.5)$ cycles: $t = 10.5 + (220/34.5) = 10.5 + 6.376 = 16.876$ cycles. At 50 Hz, 1 cycle = 20 ms, so $t = 16.876 \times 0.020 = 0.3375$ s, i.e. about 0.34 s.

Question 3:

The main advantage of single-pole auto-reclosing on an EHV line is that it

- a) is cheaper than three-pole reclosing.
- b) improves transient stability by keeping the healthy phases in service.
- c) eliminates the need for circuit breakers.
- d) increases the system fault level.

Answer: (b)

Question 4:

As per the IEC standard, a circuit breaker applied with auto-reclosing must be capable of performing the following operating cycle with full rated breaking current.

- a) O + 0.3 s + CO + 3 min + CO
- b) CO + 0.3 s + O + 3 min + O
- c) O + 3 min + CO + 0.3 s + CO
- d) O + O + O + 1 hour + O

Answer: (a)

Question 5:

Auto-reclosing is generally not employed on underground cable feeders because cable faults are usually

- a) transient in nature.

- b) self-clearing.
- c) permanent in nature.
- d) of very low magnitude.

Answer: (c)

Question 6:

In a recloser–fuse coordination scheme, the recloser is set to operate first on its fast curve so that

- a) the lateral fuse always blows first.
- b) transient faults are cleared before the lateral fuse is damaged.
- c) the recloser locks out immediately.
- d) the load current is interrupted.

Answer: (b)

Question 7:

Compared with a permissive transfer-trip scheme, the carrier-blocking scheme has the disadvantage that

- a) a deliberate coordination time delay must be allowed to wait for a possible blocking signal.
- b) it operates faster than necessary.
- c) it requires no relays.
- d) it cannot detect external faults.

Answer: (a)

Question 8:

The dead time of an auto-reclosing relay (ARR) must always be set

- a) less than the dead time of the circuit breaker.
- b) greater than the dead time of the circuit breaker.
- c) equal to the reclaim time.
- d) equal to zero.

Answer: (b)

Question 9:

The reclaim time of an auto-reclosing scheme is best defined as the

- a) time between energisation of the closing circuit and contact closure.
- b) de-ionisation time of the fault arc.
- c) arcing time of the circuit breaker.
- d) time interval between two successive (adjacent) reclosing attempts.

Answer: (d)

Question 10:

For EHV/UHV lines protected by distance relays, the auto-reclosing relay is normally permitted to operate only for faults detected in

- a) Zone-1
- b) Zone-2
- c) Zone-3
- d) all zones equally

Answer: (a)

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